

Model Comparison Matrix

Classification Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.84	0.71	0.55	0.62	0.89
Decision Tree Classifier	0.85	0.68	0.67	0.68	0.89
Random Forest Classifier	0.84	0.7	0.6	0.65	0.89
Gradient Boosting Classifier	0.86	0.76	0.6	0.67	0.92
K-Nearest Neighbors (KNN)	0.84	0.66	0.64	0.65	0.87
Support Vector Classifier (SVC)	0.8	0.72	0.28	0.4	0.84
Voting (Soft) - Top 3 Models	0.8633	-	-	-	0.92
Ensemble (Bayesian Stacking)	0.8682	-	-	-	-

Which model gave you the best result and which was the worst?

The model with the best result was Gradient Boosting Classifier with 0.86 accuracy and 0.92 ROC-AUC score. It is more trustworthy and it is the best at distinguishing between the two income classes.

However, the overall winner is Bayesian Stacking Classifier that takes predictions from Gradient Boosting, Random Forest, and Decision Trees and provides a final vote with 0.86 accuracy. It refers to other models when Gradient Boosting is unsure and takes their results into consideration.

Which model gave you the worst result and which was the worst?

SVC was the worst performer because it had very low Recall (0.28). It missed way too many high-income earners to be considered effective for this specific dataset.